Week 11
Lecture 15
Ideate, Design, Prototyping And Construction

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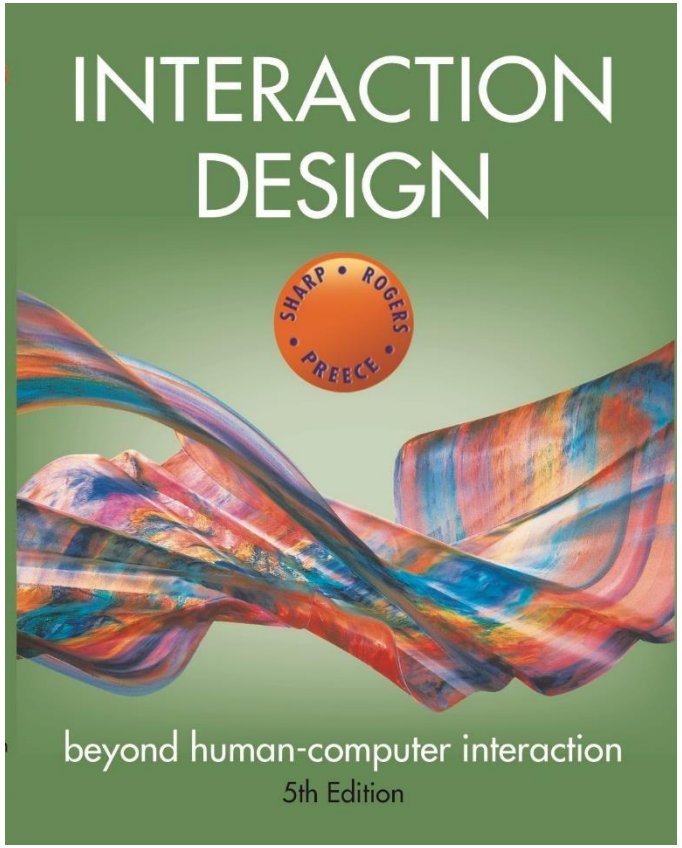

INTERACTION
DESIGN
SHARP • ROGERS • PREECE
beyond human-computer interaction
5th Edition

Chapter 12

Overview

- Prototyping
- Conceptual design
- Concrete design
- Using scenarios
- Generating prototypes
- Construction

Prototyping

- What is a prototype?
- Why prototype?
- Different kinds of prototyping
  - Low fidelity
  - High fidelity
- Compromises in prototyping
  - Vertical
  - Horizontal
- Final product needs to be engineered

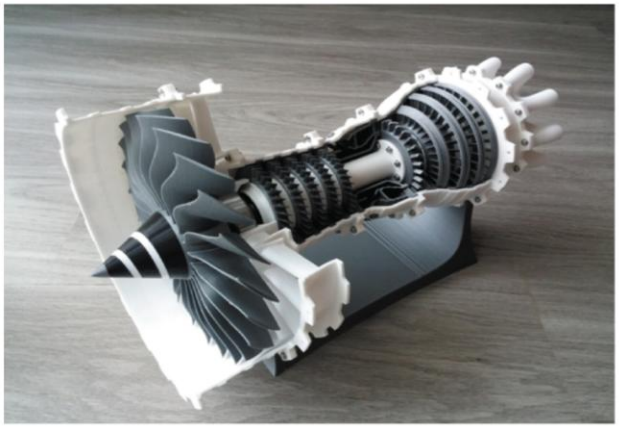
Prototyping: What is a prototype?

- One manifestation of a design that allows stakeholders to interact with it

- In other design fields, a prototype is a small-scale model:
  - A miniature car
  - A miniature building or town

Source: PalmPilot wooden model © Mark Richards

Prototyping: Examples



(a)

(b)

(c)

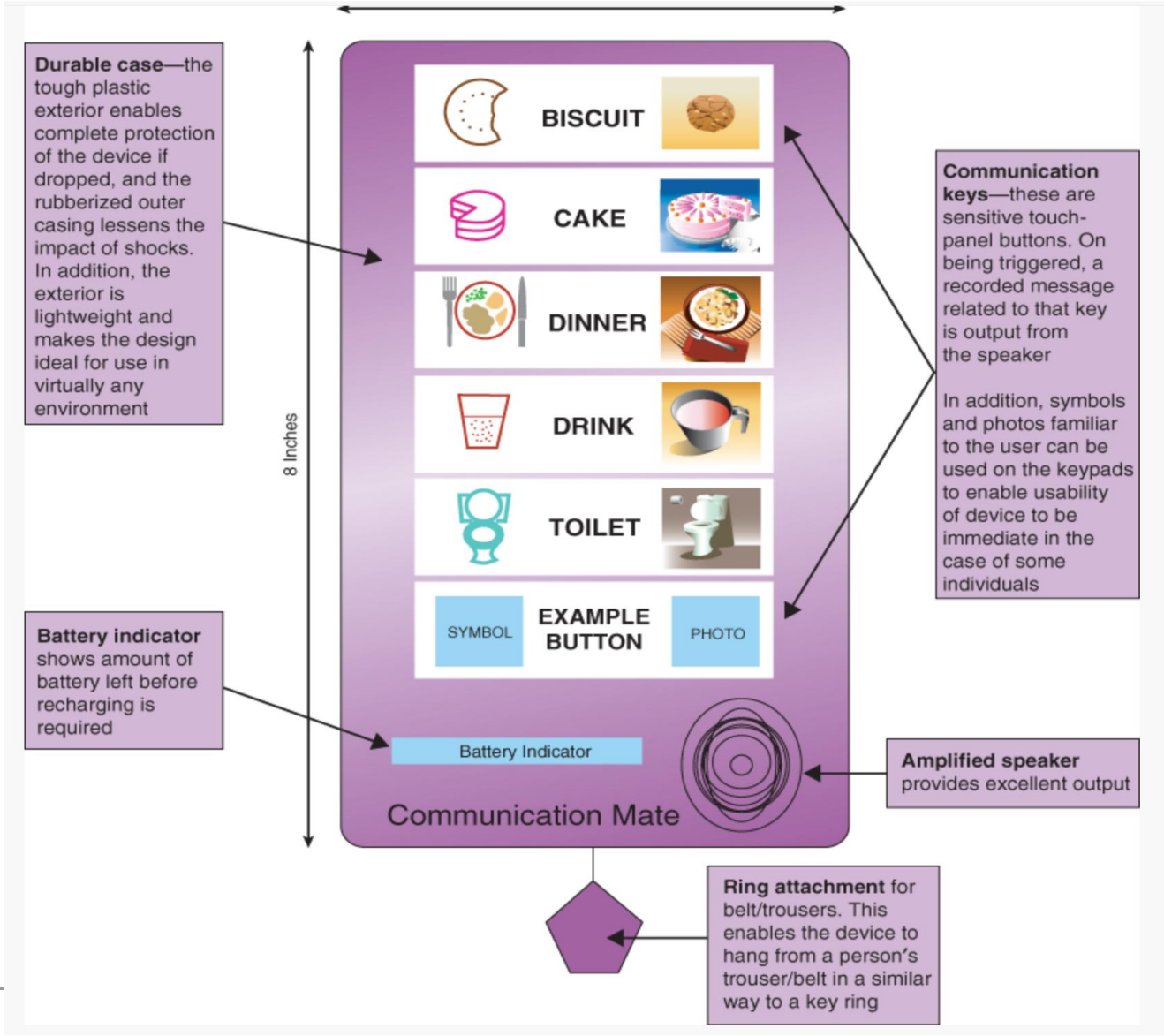
Prototyping: What is a prototype in interaction design?

- In interaction design, a prototype can be (among other things):
  - A series of **screen sketches**
  - A **storyboard**, for example, a cartoon-like series of scenes
  - A **PowerPoint slide** show
  - A **video simulating** the use of a system
  - A **lump of wood** (for instance, the PalmPilot)
  - A **cardboard** mock-up
  - A **piece of software** with limited functionality written in the target language or in another language

Prototyping: Why prototype?

- Evaluation and feedback are central to interaction design
- Stakeholders can see, hold, and interact with a prototype more easily than a document or a drawing
- Team members can communicate effectively
- Ideas can be tested out
- Prototyping encourages **reflection**: an important aspect of design
- Prototypes **answer questions** and support designers in **choosing between alternatives**

Prototyping: An Example



Prototyping: Low-fidelity (LF) Prototyping

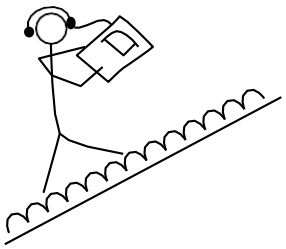
- Uses a medium which is unlike the final medium, for example, paper or cardboard
- Is quick, cheap, and easily changed
- Examples:
  - Sketches of screens, task sequences, and so on
  - ‘Post-it’ notes
  - Storyboards
  - ‘Wizard-of-Oz’

Prototyping: LF: Storyboards

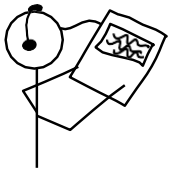
- It is a series of sketches showing how a **user might progress** through a task using the product

- Often used with scenarios, bringing in more detail and a chance to role play

Prototyping: LF: Example storyboard

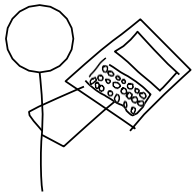


Christina walks up hill; the product gives her information about the site

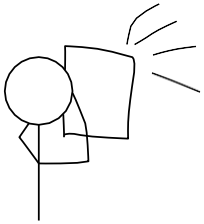


Christina adjusts the preferences to find information about the pottery trade in Ancient Greece

Christina scrambles to the highest point



Christina stores information about the pottery trader's way of life in Ancient Greece

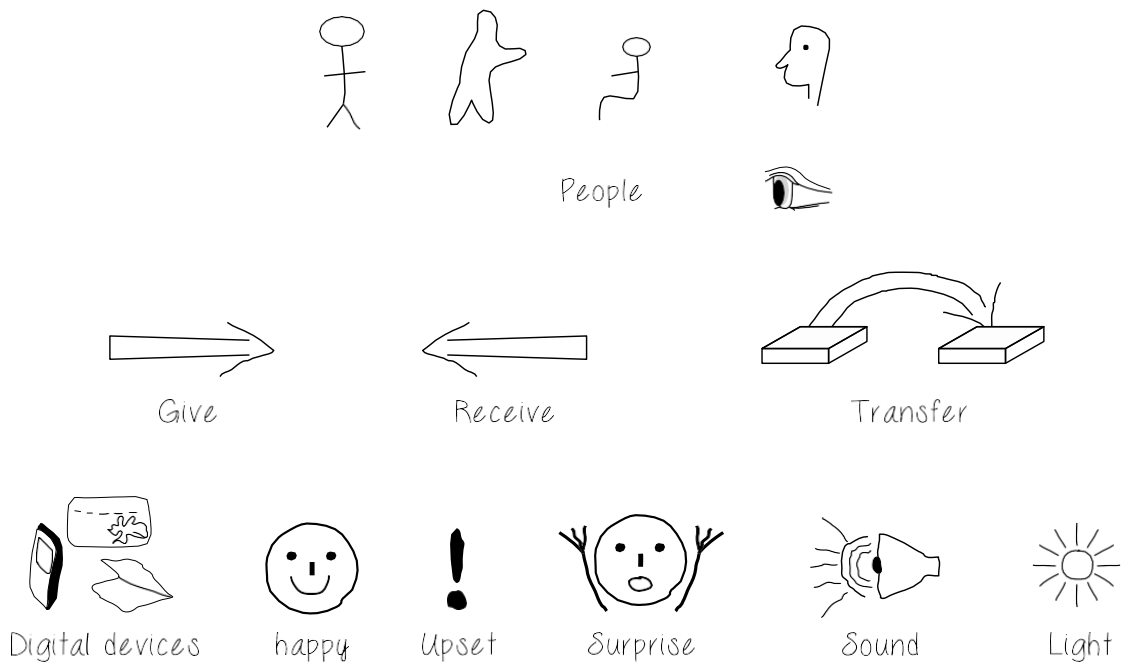


Christina takes a photograph of the location of the pottery market

Prototyping: LF: Sketching

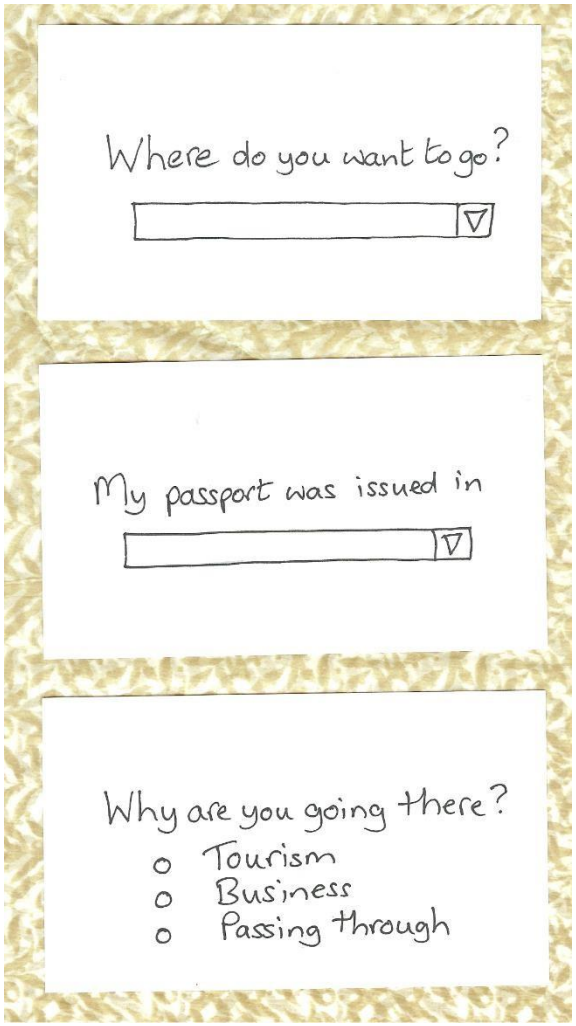
- Low-fidelity prototyping often relies on sketching

- Don't be inhibited about drawing ability — Practice simple symbols



Prototyping: LF: Index cards (1)

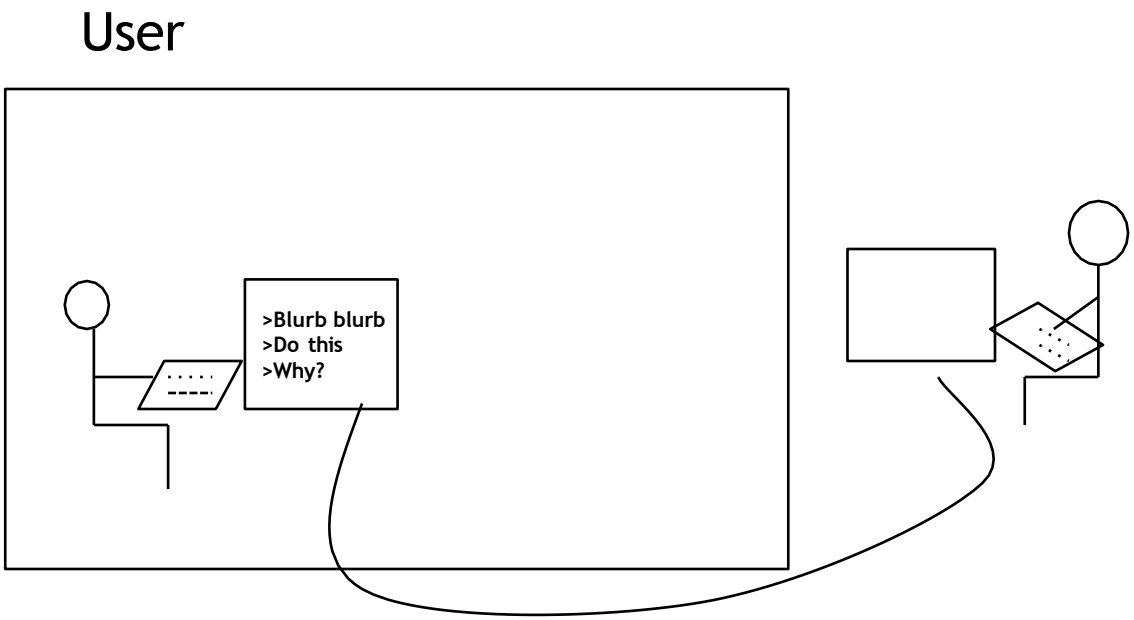
- Index cards (3 x 5 inches)
- Each card represents one element of interaction
- In evaluation, can step through the cards



Prototyping: LF: Index cards (2)

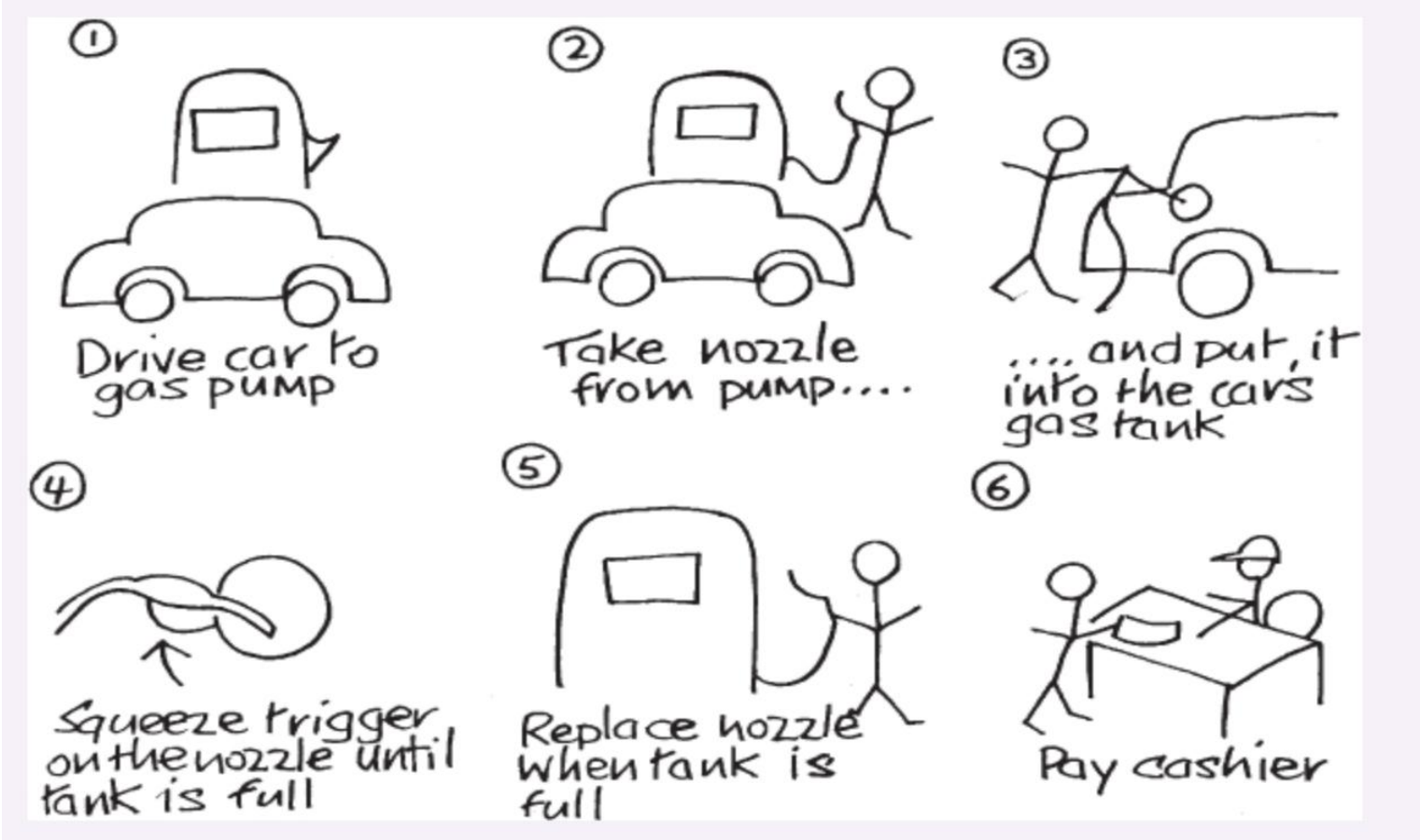
Prototyping: LF: ‘Wizard-of-Oz’

- The user thinks they are interacting with a computer, but a human is responding to output rather than the system
- Usually done early in design to understand users’ expectations



Prototyping: LF: Activity

Produce a storyboard that depicts how to fill a car with fuel.



Prototyping: High-fidelity (HF)

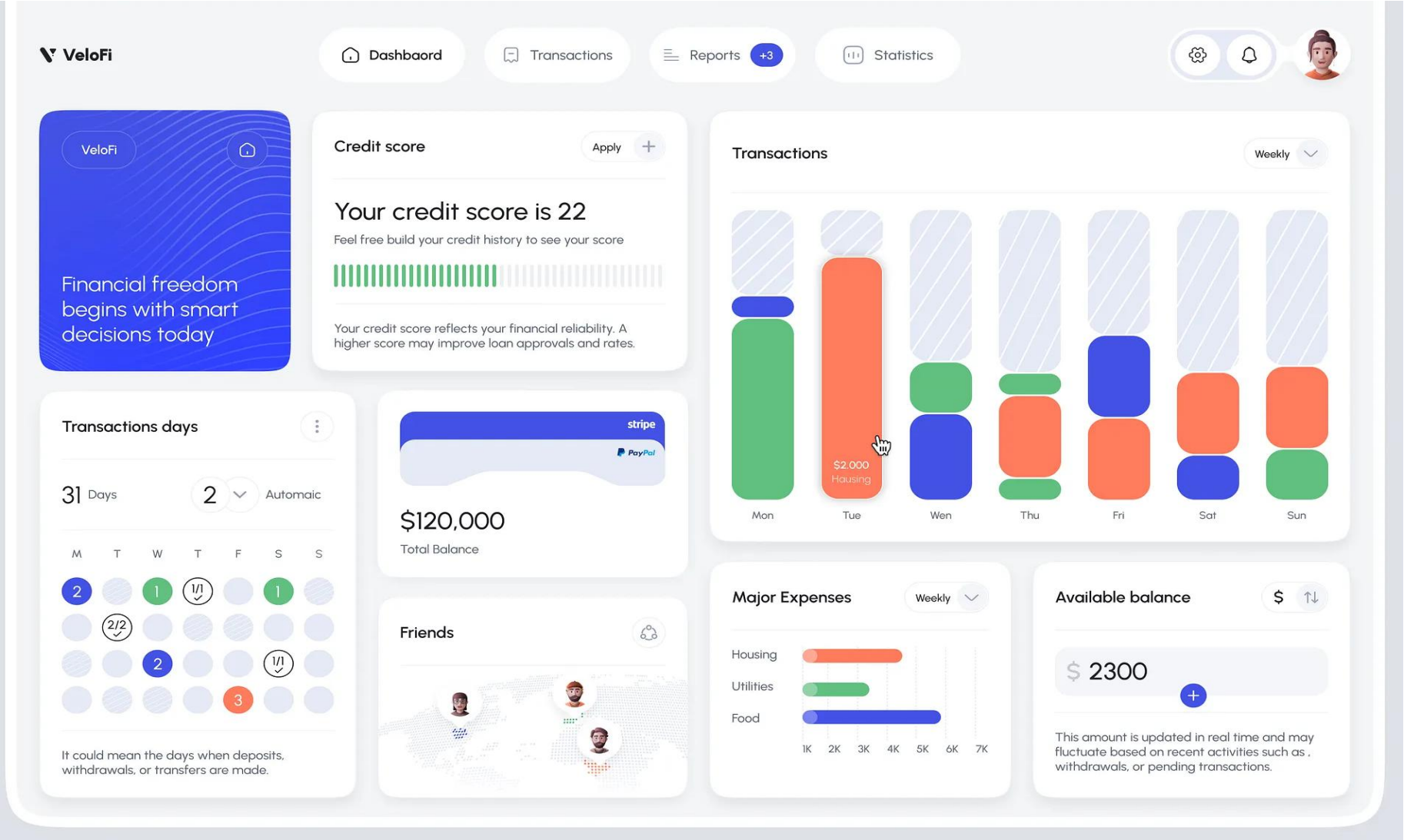
- Uses materials that you would expect to be in the final product
- Prototype looks more like the final system than a low-fidelity version
- High-fidelity prototypes can be developed by integrating existing hardware and software components
- Danger that users think they have a complete system…see compromises

Prototyping: HF: Compromises

- Prototyping involve compromises
- For software-based prototyping, maybe there is a slow response? sketchy icons? limited functionality?

- Two common types of compromise:
- **Horizontal:**
  Provides a wide range of functions, but with little detail
- **Vertical:**
  Provides a lot of detail for only a few functions
- Compromises in prototypes must not be ignored. Product needs engineering

Prototyping: HF: Example



https://dribbble.com/shots/25740162-UI-UX-Design-for-Finance-Dashboard

Prototyping: HF: Filters and Manifestations (1)

The Anatomy of Prototyping: Filters and Manifestations
Prototypes act as filters, for example, to emphasize specific aspects of a product being explored by the prototype (and to de-emphasize or omit others), and as manifestations of designs, for example, to help designers develop their design ideas through external representations (Lim et al., 2008).

Three key principles underpin the anatomy of prototypes:

Fundamental prototyping principle: Prototyping is an activity with the purpose of creating a manifestation that, in its simplest form, **filters the qualities in which designers are interested** without distorting the understanding of the whole.

Economic principle of prototyping: The best prototype is one that, in the **simplest and the most efficient way**, makes the possibilities and limitations of a design idea visible and measurable.

Anatomy of prototypes: Prototypes are filters that traverse a design space, i.e., by building different prototypes that are constrained in different ways, a designer can consider a wide range of possibilities within the design space. Prototypes are also manifestations of specific design ideas that concretize and externalize conceptual ideas.

Prototyping: HF: Filters and Manifestations (2)

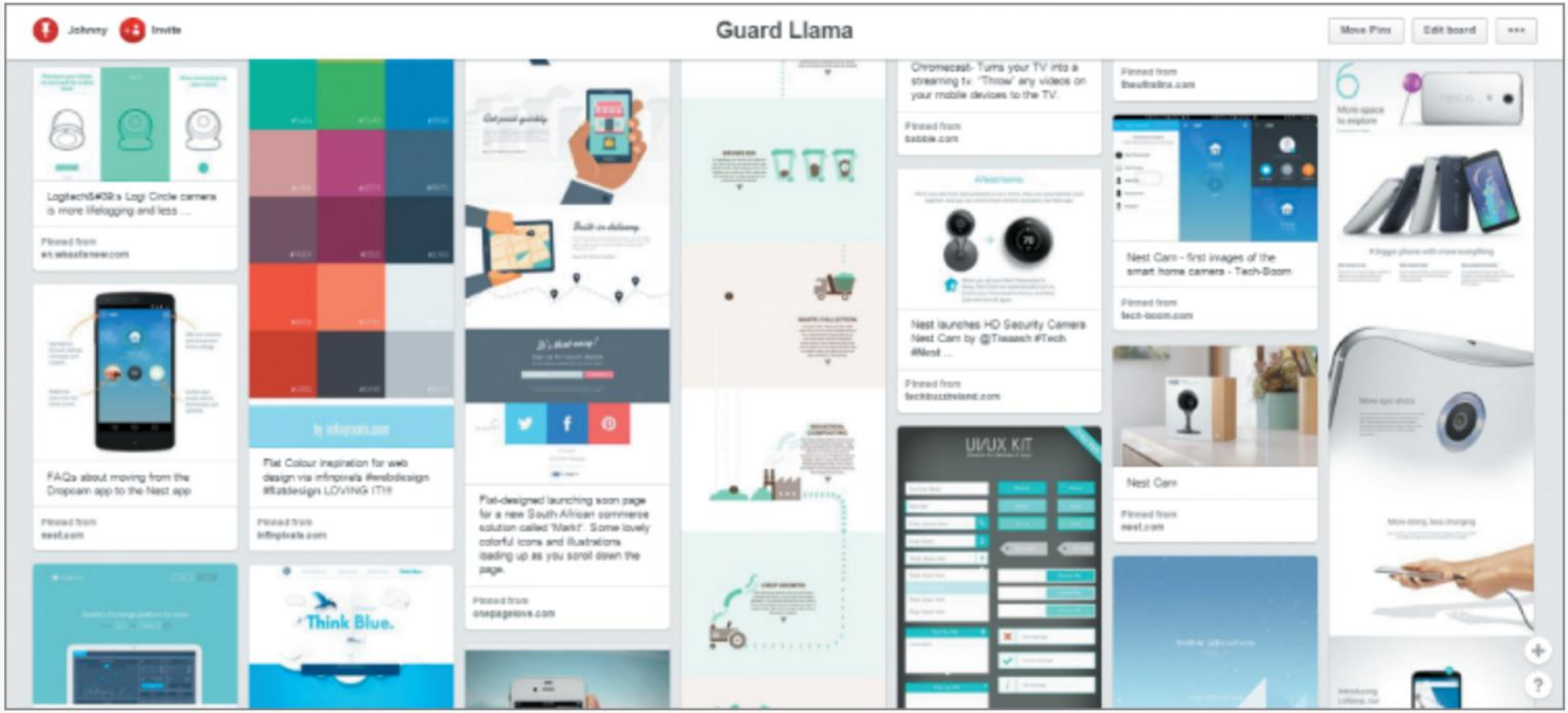
Table 12.1 Example variables of each filtering dimension

Filtering dimension	Example variables
Appearance	Size, color, shape, margin, form, weight, texture, proportion, hardness, tra
Data	Data size, data type (for example, number, string, media), data use, privac
Functionality	System function, functionality needs
Interactivity	Input behavior, output behavior, feedback behavior, information behavior
Spatial structure	Arrangement of interface or information elements; relationship among in intangible or tangible, or mixed

Conceptual design (CD)

- A conceptual model is an outline of what people can do with a product and what concepts are needed to understand and interact with it
- Understand problem space and current requirements; empathize with users
- Creativity and brainstorming techniques
- Mood board may capture desired feel
- Consider alternatives: scenarios and prototyping helps

CD: Mood Board for a personal safety product



http://johnnyhuang.design/guardllama.html

CD: Plus and minus scenarios

Scene 2: breakfast at home	
Peter starts preparing his breakfast with his new glasses on. His wife notices them and he *keenly* gives her a demonstration of what they are and how they work, and tells her about the microchip. She seems *impressed* and leaves the room to get ready for work. Peter opens the fridge to put away the butter and sees a pastry. He looks at it and gets a DietMon message telling him the calorie content of the pastry. He shows that to his wife, who is entering the kitchen and looks at him with a *smile*.	Peter prepares breakfast with his new glasses on. His wife notices them. While looking at his toast, he gets a text. His wife enquires what that is. He says it's nothing and he does not feel like having toast after all. When she questions why he becomes *tense* and *reluctantly* tells her about DietMon. *Skeptical*, she leaves the room with a sarcastic comment. Peter opens the fridge and sees a pastry. As he gives in and takes a bite, he is caught by his wife, who is entering the kitchen and looks at him with a *grin*.

CD: Building scenarios

CD: Interface metaphors (1)

- Interface metaphors combine familiar knowledge with new knowledge in a way that will help people understand the product.

- For example, consider an educational system to teach 6-year-olds mathematics.

- Tom Erickson suggests a 3-step process:

- Identify functional requirements
- Under which bits of the system will cause problems
- Generate metaphors

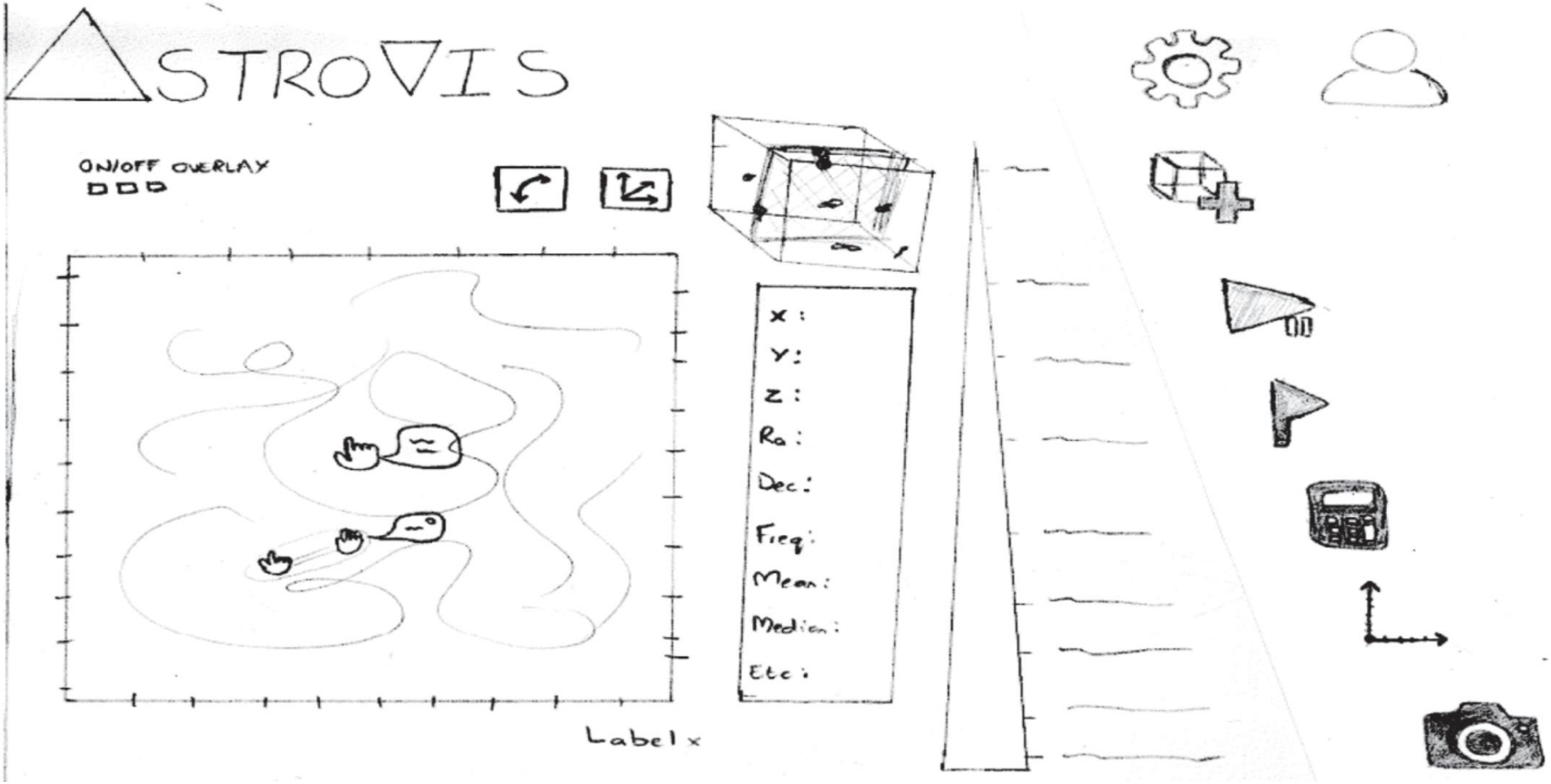
CD: Interface metaphors (2)

1. How much structure does the metaphor provide? A good metaphor will provide structure—preferably familiar structure.
2. How much of the metaphor is relevant to the problem? One of the difficulties of using metaphors is that users may think they understand more than they do and start applying inappropriate elements of the metaphor to the product, leading to confusion or false expectations.
3. Is the interface metaphor easy to represent? A good metaphor will be associated with particular physical, visual, and audio elements, as well as words.
4. Will your audience understand the metaphor?
5. How extensible is the metaphor? Does it have extra aspects that may be useful later?

Concrete design

- Difference between conceptual and concrete is emphasis
- Many aspects to concrete design
  - Color, icons, buttons, interaction devices, and so on
- Accessibility
  - Web Content Accessibility Guidelines (WCAG)
- Inclusiveness
  - Covers cultural backgrounds, language, gender, and economics

Concrete design: Example

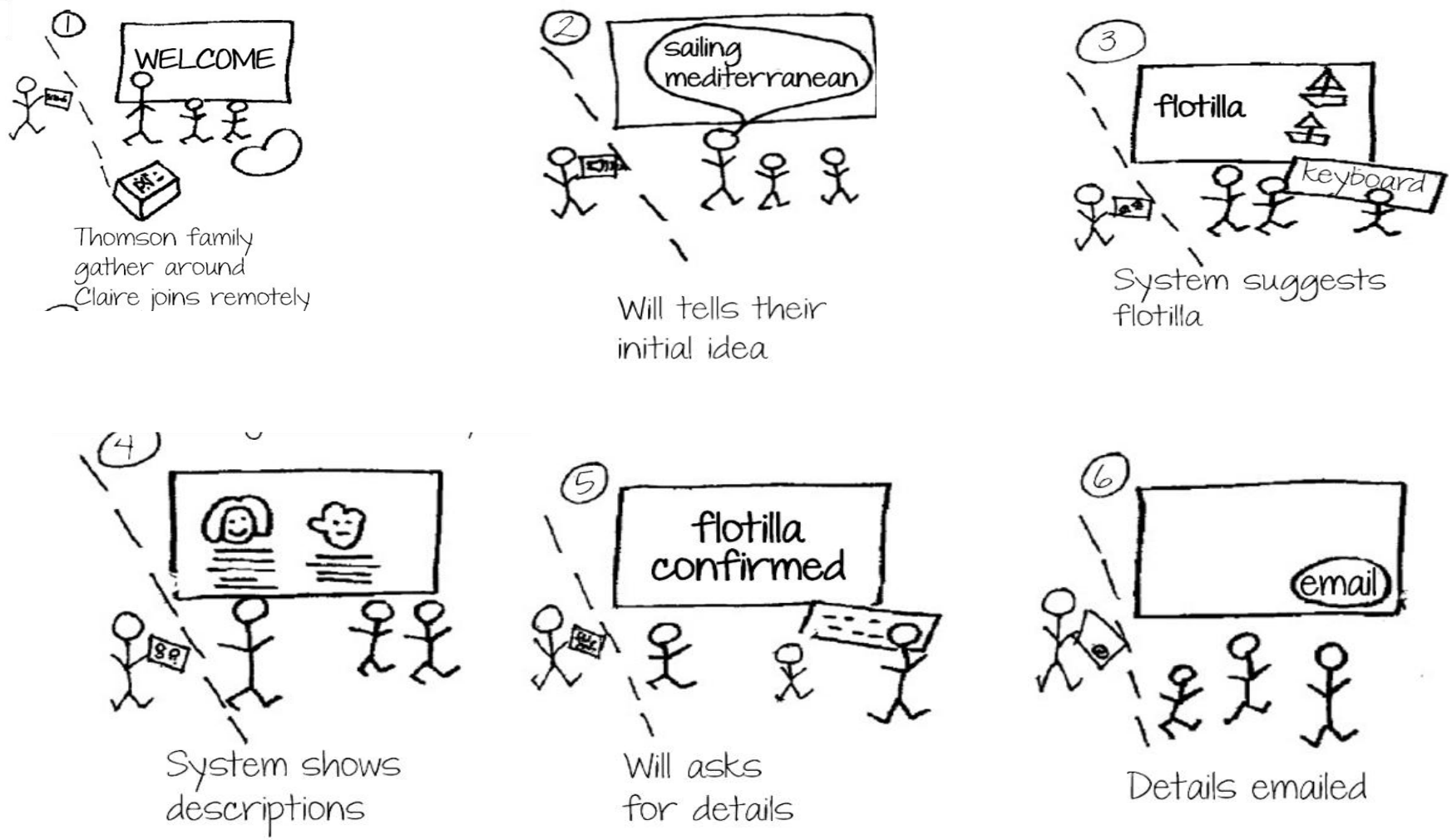


A paper prototype for the home view of an astronomy visualization package, illustrating some aspects of concrete design. On the left side is a large area for displaying the data, and on the right side there is an expanding side menu containing icons for additional functionality. In the middle is a list of data variables. The screen layout, relative sizes and locations of interface elements, and design of specific icons are all aspects of concrete design.

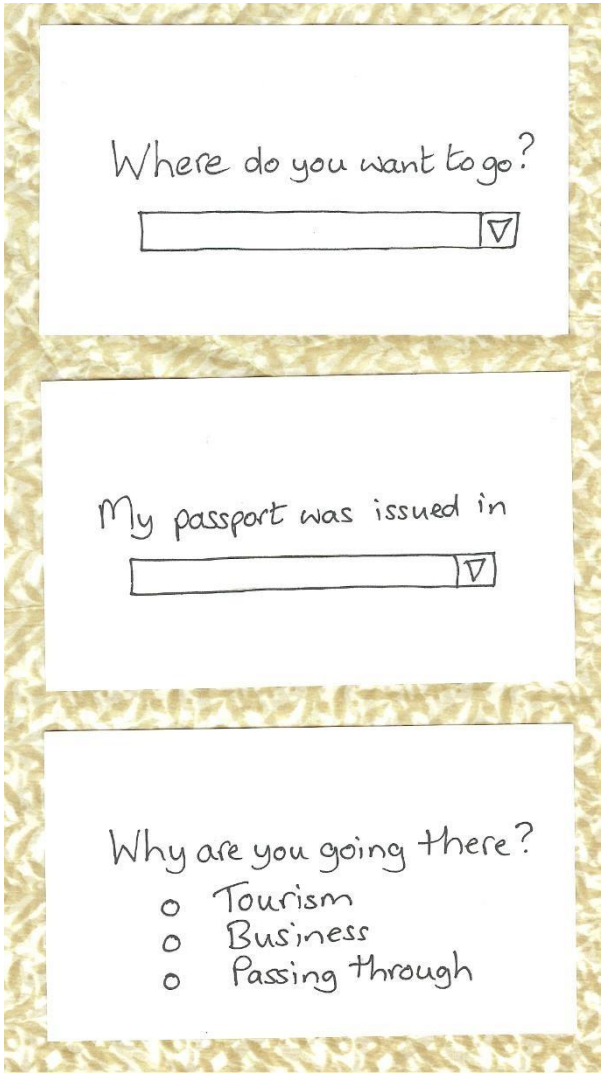
Generating prototypes

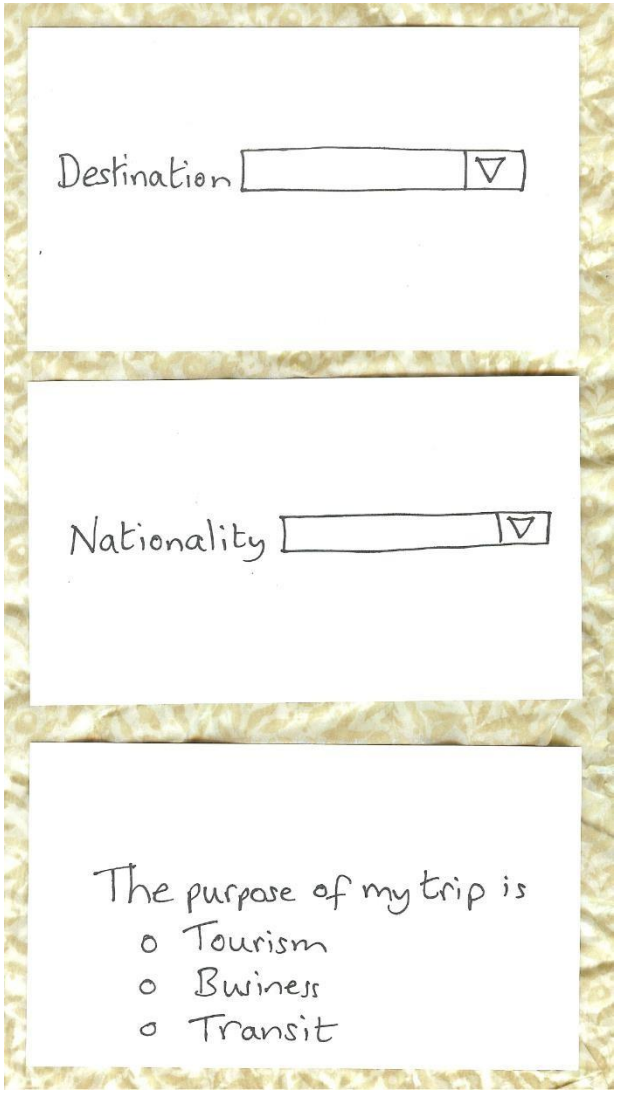
- Generate a storyboard from a scenario
  - Break down scenario into steps
  - Create a scene for each step
- Sketching out a storyboard prompts designers to think about design issues
- Generate a card-based prototype from a storyboard or from a use case
  - Consider each step in use case – what interaction element is needed
  - Draw a card that captures it

Generating prototypes: Travel Organizer (Storyboard)



Generating prototypes: Travel Organizer (Card-Based)

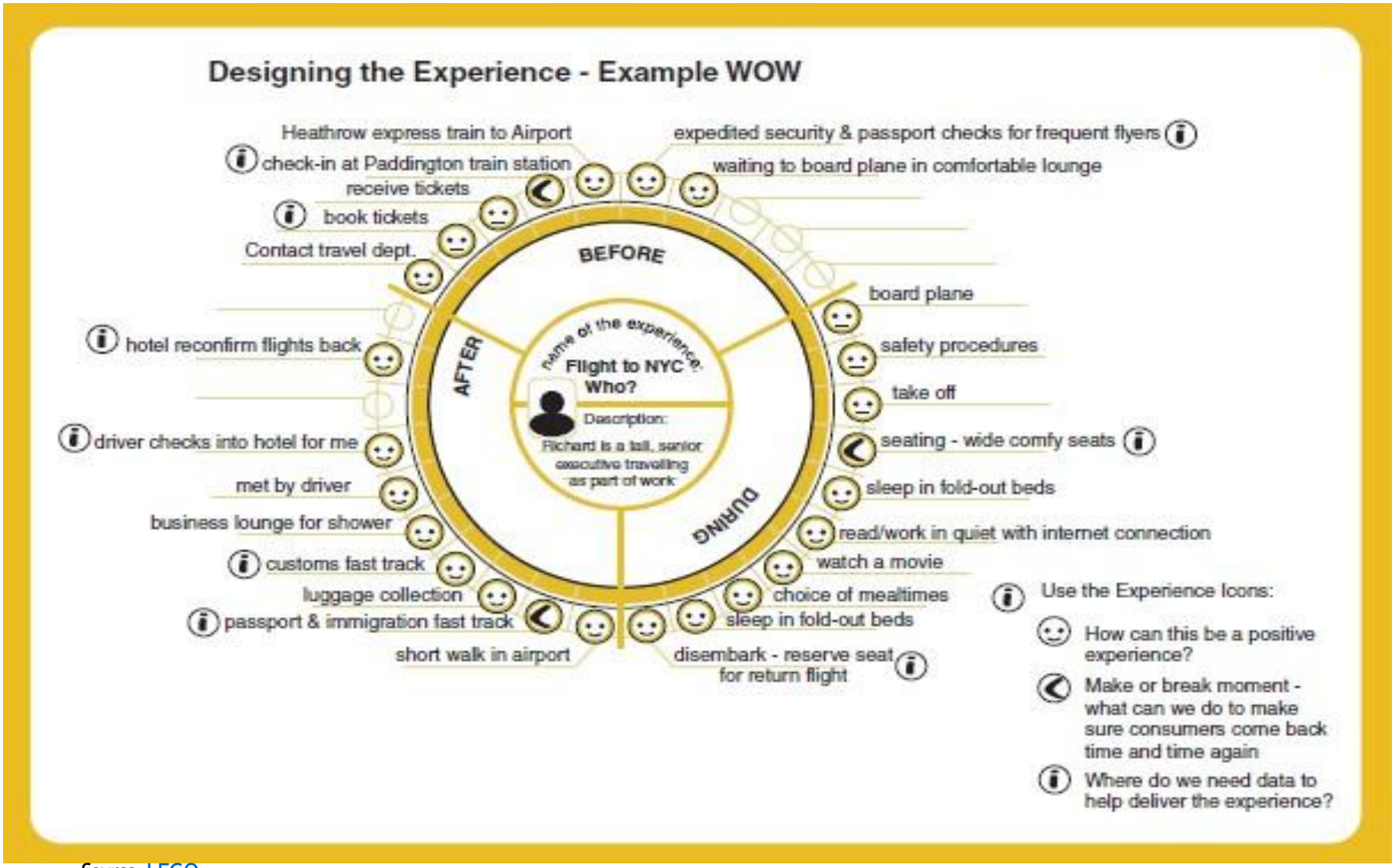




Generating prototypes: Explore the user's experience

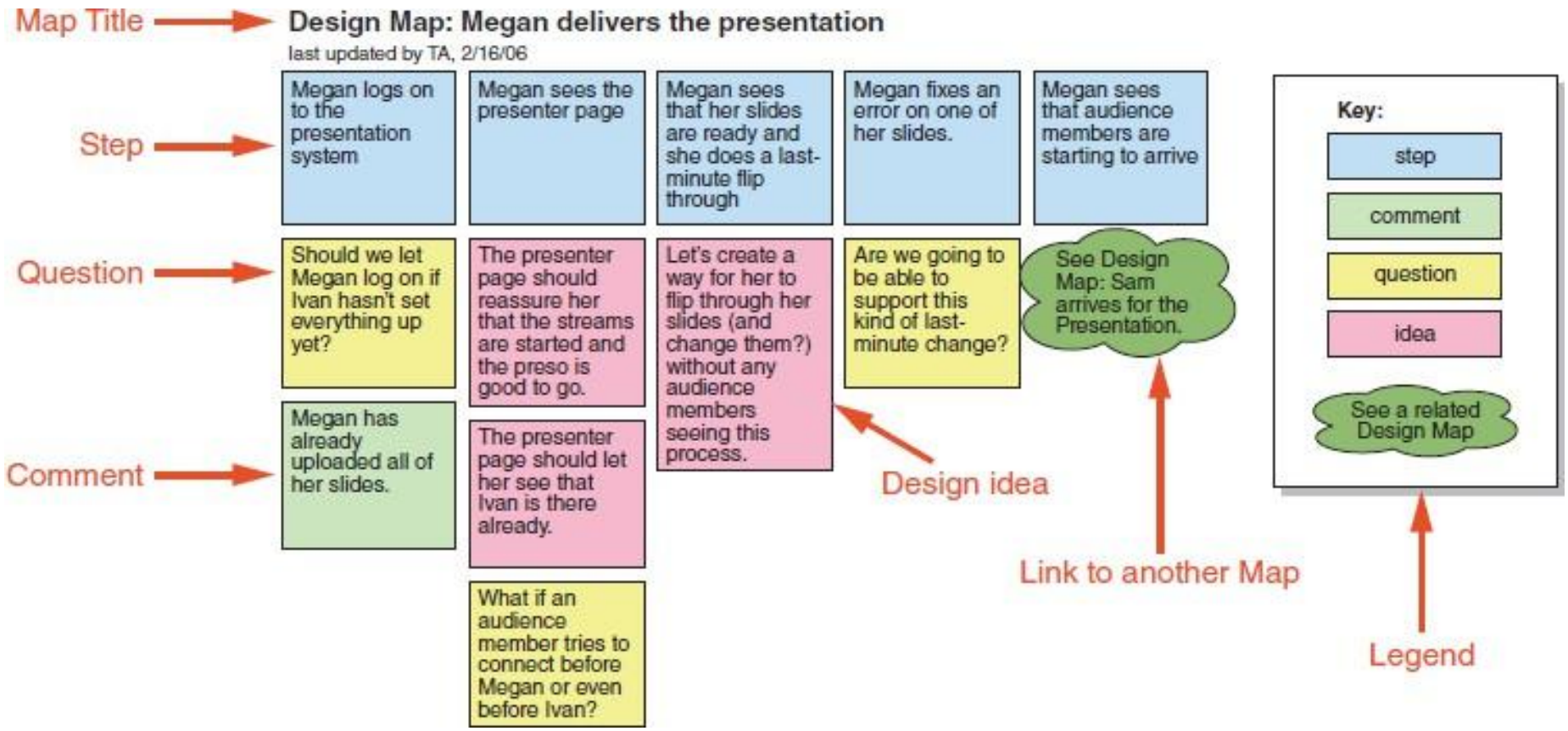
- Use personas, card-based prototypes, or stickies to model the user experience
- Visual representation called:
  - Design map
  - Customer or user journey map
  - Experience map
- Two common representations
  - Wheel
  - Timeline

Generating prototypes: Explore the user's experience – Wheel (experience map)



Source: LEGO

Generating prototypes: Explore the user's experience – Timeline (experience map)

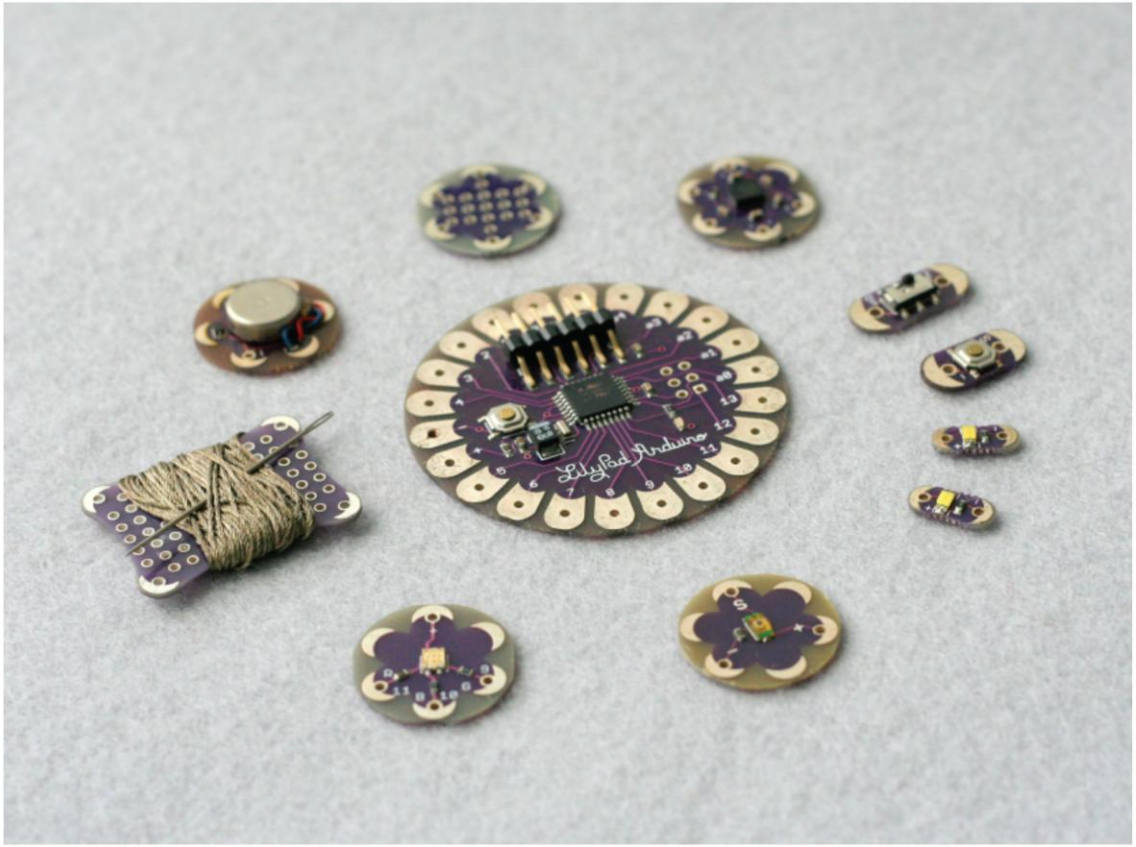


Source: Adlin and Pruitt (2010), p134. Used courtesy of Morgan Kaufmann.

Construction: SDKs

- Software Development Kits
  - Programming tools and components to develop for a specific platforms, for example, iOS
- Includes: IDE, documentation, drivers, sample code, and application programming interfaces (APIs)
- Makes development much easier
- Examples:
  - Microsoft's Kinect SDK for motion tracking
  - Building voice-based services with Amazon's Alexa Skills Kit: developer.amazon.com/alexa-skills-kit.
  - Constructing augmented reality experiences with Apple's ARKit: developer.apple.com/arkit.

Construction: Physical computing



The Lilypad Arduino kit

Summary

- Prototyping may be low fidelity (such as paper-based) or high fidelity (such as software-based)
- Existing software and hardware helps create prototypes
- Two aspects to design: conceptual and concrete
- Conceptual design develops an outline of what users can do and what concepts are needed to understand the product.
- Concrete design specifies design details, for example, layout or navigation